

Vidvathama Ramesh

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Education

International Institute of Information Technology, Hyderabad (IIIT-H) July 2024 – May 2028
B.Tech + MS (Dual Degree), Electronics and Communication Engineering; CGPA: 9.01/10.0 (9.1+ in the last three semesters)

- **Relevant Coursework:** Digital VLSI Design, Processor Architecture, Digital Systems and Logic, Operating Systems, Signal Processing, Information Theory, Data Structures and Algorithms, Statistical Methods in AI, Natural Language Processing, Systems Thinking.

Technical Projects

Adaptive Semantic Parallelism — Select-Then-Route LLM Inference 🐍 *PyTorch, DistilBERT, LightGBM, spaCy* Jan 2026 – April 2026

- Architected an 8-stage pipeline decomposing a prompt into a semantic DAG and routing each sub-task to a strong or weak model via a 27-feature LightGBM router with BFS strong-inheritance (85.66% accuracy, 88.04% F1), giving a **49% inference-cost reduction** at 131 ms mean latency.
- Fine-tuned a dual-head DistilBERT (11-class intent classification and complexity regression; 83.32% accuracy) as the front-end selector, behind an asynchronous DAG engine with speculative execution, semantic caching and a shared context blackboard.

32-bit RISC-V Processor 🐍 *Verilog, Icarus Verilog, Python, RISC-V Assembly* Feb 2025 – April 2025

- Designed a 32-bit RISC-V processor in Verilog RTL, scaling a single-cycle datapath into a 5-stage IF/ID/EX/MEM/WB pipeline: program counter, register file, instruction and data memories, ALU and control, and immediate-generation units for R-, I-, S- and B-type instructions.
- Engineered forwarding and hazard-detection units covering data, double-data, load-store, control and load-use hazards through forwarding paths, stalls and branch flushes; wrote a Python assembler and Icarus Verilog regression testbenches validating Fibonacci, linear search and vector-addition programs.

PolyQuery — Multi-Model Relational Algebra Engine 🐍 *C++17, Systems Programming, Concurrency* Sep 2025 – Nov 2025

- Engineered an integer-optimised C++17 engine unifying relational algebra, graph processing and high-dimensional vector search over a custom block-based buffer manager with external-memory algorithms (external merge sort, hashing, B-Tree indexing) under a fixed 10-page memory ceiling.
- Implemented multi-threaded transactions with Wait-Die concurrency control and page-granular shared/exclusive locking (deadlock-free), disk-backed BFS path queries and KNN search (Euclidean and cosine) via a bounded min-heap.

POSIX-Core C Shell 🐍 *C, Unix Syscalls, Signals, Valgrind* July 2025 – Aug 2025

- Developed a modular execvp-based Unix shell with foreground/background job control, unlimited pipelines, full I/O redirection and .myshrc alias resolution; verified leak-free execution under Valgrind.
- Implemented 12+ built-ins (hop, reveal, seek, proclore, activities, fg/bg, neonate and an iMan TCP man-page fetcher) with robust SIGINT/SIGTSTP/SIGCHLD handling and asynchronous child reaping.

Research Experience

Research Intern — TCS Research *Mentor: Jeripothula Prudviraj, Hyderabad* May 2026 – July 2026

- Architected deep neural encoders extracting aesthetic and stylistic features as state representations for reinforcement-learning agents in a human-AI co-creation framework where generative agents act under an inference-time RL layer (best-of-N search, batched reward model, UCB1 bandit, Reflexion self-correction).
- Submitted to **NeurIPS 2026** (Creative AI Track: Agency); **ACM CHI 2026** manuscript in preparation.

Undergraduate Researcher — Robotics Research Centre (RRC), IIIT-H *Supervisor: Prof. Sourav Garg* May 2025 – Present

- Benchmarking feedforward 3D mapping foundation models against an in-house visual navigation stack under a common output schema; isolated and mitigated metric-scale ambiguity and camera frame-convention mismatch.

Research Intern — Indian Institute of Technology Bombay *Supervisor: Prof. Vikram Bansal, Mumbai* May 2025 – Aug. 2025

- Implemented a TransCrowd transformer pipeline for high-density crowd counting, integrating spatio-temporal features to improve density-estimation accuracy.

Technical Skills

Languages: C, C++17, Python, Verilog, RISC-V Assembly, x86 Assembly, Bash, SQL, MATLAB, LaTeX

Digital Design and VLSI: RTL design, pipelining, hazard and forwarding logic, FSM design, combinational and sequential logic, static CMOS gates, transistor sizing, full-custom layout (MAGIC), NGSPICE, parasitic extraction, post-layout timing and power analysis, Icarus Verilog/GTKWave, FPGA prototyping

Systems: Linux/Unix, POSIX, system calls, signals, process management, concurrency, multithreading, parallel computing, external-memory algorithms, CPU architecture

ML and Tools: PyTorch, HuggingFace, LightGBM, Scikit-Learn, NumPy, OpenCV, Git, Docker, Valgrind, GDB, ROS 2, SQL

Achievements and Honours

- **Dean's List for Academic Excellence**, IIIT Hyderabad — top 5% of batch (Spring, UG2), top 10% of batch (Monsoon, UG3).
- **Placed 5th out of 150, Lloyds IP&I Agentic AI Hackathon 2025** — voice-first IVR replacement: Whisper speech-to-text into a RAG agent (GPT-4o over a FAISS index of policy documents) that grounds every response in source policy text and auto-files tickets.
- **1st Prize, SUD Life Bank Hackathon** — intelligent insurance platform (FinTech/banking track).